

Inversion Recovery

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1 Inversion Recovery

The Bloch equation says that longitudinal magnetization recovers from an initial condition $M_z(0)$ toward a thermal equilibrium value of M_0 as follows:

$$M_z(t) = M_z(0)e^{-t/T_1} + M_0(1 - e^{-t/T_1}) \quad (1)$$

For perfect inversion, $M_z(0) = -M_0$:

$$M_z(t) = -M_0e^{-t/T_1} + M_0(1 - e^{-t/T_1}) \quad (2)$$

$$M_z(t) = M_0(1 - 2e^{-t/T_1}) \quad (3)$$

For inversion recovery experiments, we typically call the time after an inversion pulse TI :

$$M_z(TI) = M_0(1 - 2e^{-TI/T_1}) \quad (4)$$

2 Signal Nulling

At what TI will spins with a particular inversion time have no longitudinal magnetization? Set $M_z(TI) = 0$ and solve for TI :

$$M_z(TI) = M_0(1 - 2e^{-TI/T_1}) = 0 \quad (5)$$

$$1 - 2e^{-TI/T_1} = 0 \quad (6)$$

$$-2e^{-TI/T_1} = -1 \quad (7)$$

$$e^{-TI/T_1} = \frac{1}{2} \quad (8)$$

$$\ln e^{-TI/T_1} = \ln \frac{1}{2} \quad (9)$$

$$\frac{-TI}{T_1} = \ln \frac{1}{2} \quad (10)$$

$$\frac{-TI}{T_1} = \ln 1 - \ln 2 \quad (11)$$

Since $\ln 1 = 0$:

$$\frac{-TI}{T_1} = -\ln 2 \quad (12)$$

$$TI = T_1 \ln 2 \quad (13)$$